Q1:

Argue: Consider starting from i. $\forall j \neq i$ by the definition of irreducible, $\exists n > 0$ s.t. $P_{ij}^n > 0$.

Since each state is recurrent i will come back to i with probability 1, which implies that each chain can always try to attain j with positive possibility everytime starting from i. Thus the chain will go to every other state with possibility 1

Q2:

the X_{i+1} only depends on X_i. if $X_i=0$, $P(X_{i+1}=1|X_i=0)=\lambda(1-\mu)$, $P(X_{i+1}=0|X_i=0)=1-\lambda+\lambda\mu$

if $X_i>0$, $P(X_{i+1}=k_i+1|X_i=k_i)=(1-\lambda)\mu$

$P(X_{i+1}=k_i|X_i=k_i)=\lambda\mu+(1-\lambda)(1-\mu)=2\lambda\mu-\lambda-\mu+1$

$P(X_{i+1}=k_i+1|X_i=k_i)=\lambda(1-\mu)$

$$\therefore \begin{pmatrix} 1-\lambda+\lambda\mu & \lambda(1-\mu) & 0 & \\ \mu(1-\lambda) & 1-\lambda-\mu+2\lambda\mu & \lambda(1-\mu) & 0 \\ 0 & \mu(1-\lambda) & 1-\lambda-\mu+2\lambda\mu & \lambda(1-\mu) \\ 0 & 0 & \ddots & \ddots \end{pmatrix}$$

And each state communicate with each other state ∴ irreducible.

(2): the Y_{i+1} only depends on Y_i, if $Y_i=0$, $P(Y_{i+1}=1)=1$

if $Y_i\neq 0$, $P(Y_{i+1}=k+1|Y_i=k)=P(\text{X change to}|\text{the X change})=\frac{P(\text{X add 1})}{P(\text{X add 1 or X decrease})}$

$$=\frac{\lambda(1-\mu)}{\lambda(1-\mu)+(1-\lambda)\mu}=\frac{\lambda(1-\mu)}{\lambda+\mu-2\lambda\mu}$$

$P(Y_{i+1}=k+1|Y_i=k)=\frac{\mu(1-\lambda)}{\lambda+\mu-2\lambda\mu}$

$$\therefore \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ \frac{\mu(1-\lambda)}{\lambda+\mu-2\lambda\mu} & 0 & \frac{\lambda(1-\mu)}{\lambda+\mu-2\lambda\mu} & 0 & 0 \\ 0 & \frac{\mu(1-\lambda)}{\lambda+\mu-2\lambda\mu} & 0 & \frac{\lambda(1-\mu)}{\lambda+\mu-2\lambda\mu} & 0 \\ & \ddots & \ddots & \ddots & \ddots \end{pmatrix}$$

(3):

①: Start from 0, the next state will always be 1

and start from 1, it is a "gambler's game"

$$P(\text{start from 1 and return to 0}) = \begin{cases} 1-\frac{1-(\frac{q}{p})}{1-(\frac{q}{p})^N} & p \in (0,1)\setminus\{\frac{1}{2}\} \\ 1-\frac{1}{N} & p=\frac{1}{2} \\ 1 & p=0 \\ 0 & p=1 \end{cases}$$

$$\lim_{N\to\infty} P(\text{start from 1 and return to 0}) = 1 \quad q=1, \text{ or } \frac{q}{p}>1, \text{ or } q=\frac{1}{2}$$

i.e: $\frac{1}{\sum \lambda+M-2\lambda M} \le 1 \Rightarrow M \ge \lambda$

d) just the same

If recurrent

$M(1-\lambda) \ge \lambda(1-M) \quad M \ge \lambda$

If transient: $\lambda > M$

e): $\pi = \pi P, \ \pi \mathbf{1} = 1$.

$\pi_0 = \pi_0(1-(1-M)\lambda) + \pi_1(1-\lambda)M \Rightarrow \frac{\pi_1}{\pi_0} = \frac{\lambda(1-M)}{M(1-\lambda)}$

$\pi_1 = \pi_0 \lambda(1-M) + \pi_1(1-M-\lambda+2M\lambda) + \pi_2(1-\lambda)M \Rightarrow \frac{\pi_2}{\pi_1} = \frac{\lambda(1-M)}{M(1-\lambda)}$

$\frac{\pi_{i+1}}{\pi_i} = t$.

$\pi_0(1+t+t^2+\cdots) = 1, \ \pi_0 \frac{1}{1-t} = 1$

$\therefore \pi_0 = 1-t$

$\therefore \pi = (1-t, (1-t)t, \cdots) = \left(1-\frac{\lambda(1-M)}{M(1-\lambda)}, \left(1-\frac{\lambda(1-M)}{M(1-\lambda)}\right)\left(\frac{\lambda(1-M)}{M(1-\lambda)}\right), \left(1-\frac{\lambda(1-M)}{M(1-\lambda)}\right)\left(\frac{\lambda(1-M)}{M(1-\lambda)}\right)^2, \cdots\right)$

Q3:

Given the robot move away from i, the possibility it moves to i+1 mod (n+1) is p, to (i-1) mod (n+1) is 1-p

so it has p: move to 1, and (1-p) move to n.

① it move to 1, it means that it is like a gambler who has "one" dollar, he need to win a person have (i-1) dollar, each time he has "p" possibility to win

so we consider p_1 in $N=i$

we know $P_0=0$, $P_i=0$, $P_k=pP_{k+1}+qP_{k-1}$ $q=1-p$

Guess $P_i=x^i$, $x^i=px^{i+1}+qx^{i-1}$, $x=\frac{1}{2p}(1\pm\sqrt{1-4pq})$ $\in\{1, \frac{q}{p}\}$

$P_i=A1^i+B(\frac{q}{p})^i$ $(p\neq q)$

$P_0=0\Rightarrow B=-A$, $P_N=1\Rightarrow A(1-(\frac{q}{p})^N)=1$,

$\therefore P_i=\begin{cases}\frac{1-(\frac{q}{p})^i}{1-(\frac{q}{p})^N} & \text{if } p\neq q.\\ \frac{i}{N} & \text{if } p=q.\end{cases}$

∴ we consider in the problem set. $N=i$ and $i=1$

$P_1=\frac{1-\frac{q}{p}}{1-(\frac{q}{p})^i}$

∴ the first probability is $\frac{p-q}{1-(\frac{q}{p})^i}$ (if $p=\frac{1}{2}$, then is $\frac{p}{i}$)

the second is just the same process: set $N=n-i+1$

the second probability is $q\cdot\frac{1-\frac{p}{q}}{1-(\frac{p}{q})^{n-i+1}}=\frac{q-p}{1-(\frac{p}{q})^{n-i+1}}$ (if $p=\frac{1}{2}$, then is $\frac{q}{n-i+1}$)

Above all

$$\begin{cases}\frac{1}{2}(\frac{1}{i}+\frac{1}{n-i+1}) & (p=\frac{1}{2})\\ \frac{2p-1}{1-(\frac{1-p}{p})^i}+\frac{1-2p}{1-(\frac{p}{1-p})^{n-i+1}} & (p\in(0,1)\setminus\{\frac{1}{2}\})\\ 1 & (p=0 \text{ or } 1) \text{ because it move along one direction}\end{cases}$$

b)

It move to 1 first, set $i=n$ $\quad P_1 = \begin{cases} \frac{2p-1}{1-(\frac{1-p}{p})^n} & p\in(0,1)\ \&\ p\neq\frac{1}{2} \\ \frac{p}{n} & p=\frac{1}{2} \\ p & p=0 \text{ or } 1 \end{cases}$

It move to 2 first set $i=1$: $\quad P_2 = \begin{cases} \frac{1-2p}{1-(\frac{p}{1-p})^n} & (p\in(0,1)\ \&\ p\neq\frac{1}{2}) \\ \frac{1-p}{n} & p=\frac{1}{2} \\ 1-p & p=0 \text{ or } 1 \end{cases}$

$$\therefore P_{all} = P_1 + P_2 = \begin{cases} \frac{2p-1}{1-(\frac{1-p}{p})^n} + \frac{1-2p}{1-(\frac{p}{1-p})^n} & (p\in(0,1)\ \&\ p\neq\frac{1}{2}) \\ \frac{1}{n} & (p=\frac{1}{2}) \\ 1 & (p=0 \text{ or } 1) \end{cases}$$

(c): Given that it move back after inspect all point, what's possibility it move to 1 firstly?

① $p=\frac{1}{2}$, certainly the possibility is $\frac{1}{2}$

② $p=0$. It's impossible to move to 1 first $\quad P'=0$

③ $p=1$ It certainly moves to 1 first $\quad P'=1$

④ $p\in(0,1)$ P(move to 1 first | come back before inspecting all)

$$= \frac{P(\text{move to 1 first, come back before inspecting all})}{P(\text{come back before inspecting all})} = \frac{\frac{2p-1}{1-(\frac{1-p}{p})^n}}{\frac{2p-1}{1-(\frac{1-p}{p})^n} + \frac{1-2p}{1-(\frac{p}{1-p})^n}} = \frac{1}{1+(\frac{1-p}{p})^n}$$

$$\therefore P' = \begin{cases} \frac{1}{2} & p=\frac{1}{2} \\ 0 & p=0 \\ 1 & p=1 \\ \frac{1}{1+(\frac{1-p}{p})^n} & p\in(0,1)\setminus\{\frac{1}{2}\} \end{cases}$$

Q4

Let X_i denote the number of people in ith generation, $X_1 = n$

$E(X_{n+1}) = E\{E(X_{n+1}|X_n)\} = E(mX_n) = mE(X_n)$

$\therefore E(X_n) = nm^{n-1}$

$E\left(\sum_{i=1}^{\infty} X_i\right) = \sum_{i=1}^{\infty} E(X_i) = \sum_{i=1}^{\infty} nm^{i-1} = \lim_{k\to\infty} \frac{1(1-m^k)}{1-m} \cdot n = \frac{n}{1-m}$

$\therefore \frac{n}{1-m}$